MED-35

35 ft Composite Cruiser-Racer

A performance-oriented sailing yacht designed for family cruising in the Mediterranean. Balancing speed in light conditions with safety and ease of handling for short-handed crews.

DESIGNER

Roman Aime

INSTITUTION

Solent University

MODULE

SEF533 — Sailing Yacht Design

DATE

January 2026

01

Design Philosophy

The MED-35 has been optimised for family cruising in the Mediterranean, with a cruiser-racer performance target that prioritises a balanced hull capable of producing speed in light wind conditions while remaining tolerant and safe as the wind increases.

The LWL/LOA ratio was chosen just above average to maximise waterline length and maintain regular boat speed in light conditions. The LWL/BWL ratio of 3.25 and Slenderness Ratio of 5.6 ensure a sufficiently slender hull to limit drag without sacrificing internal volume.

The SA/Displacement ratio of 20 is positioned below the trend to keep sail handling simple and remain consistent with a family cruising philosophy, while the Ballast Ratio of 0.33 avoids an excessively stiff configuration. The Cp value of 0.57 balances speed and stability for an efficient hull form.

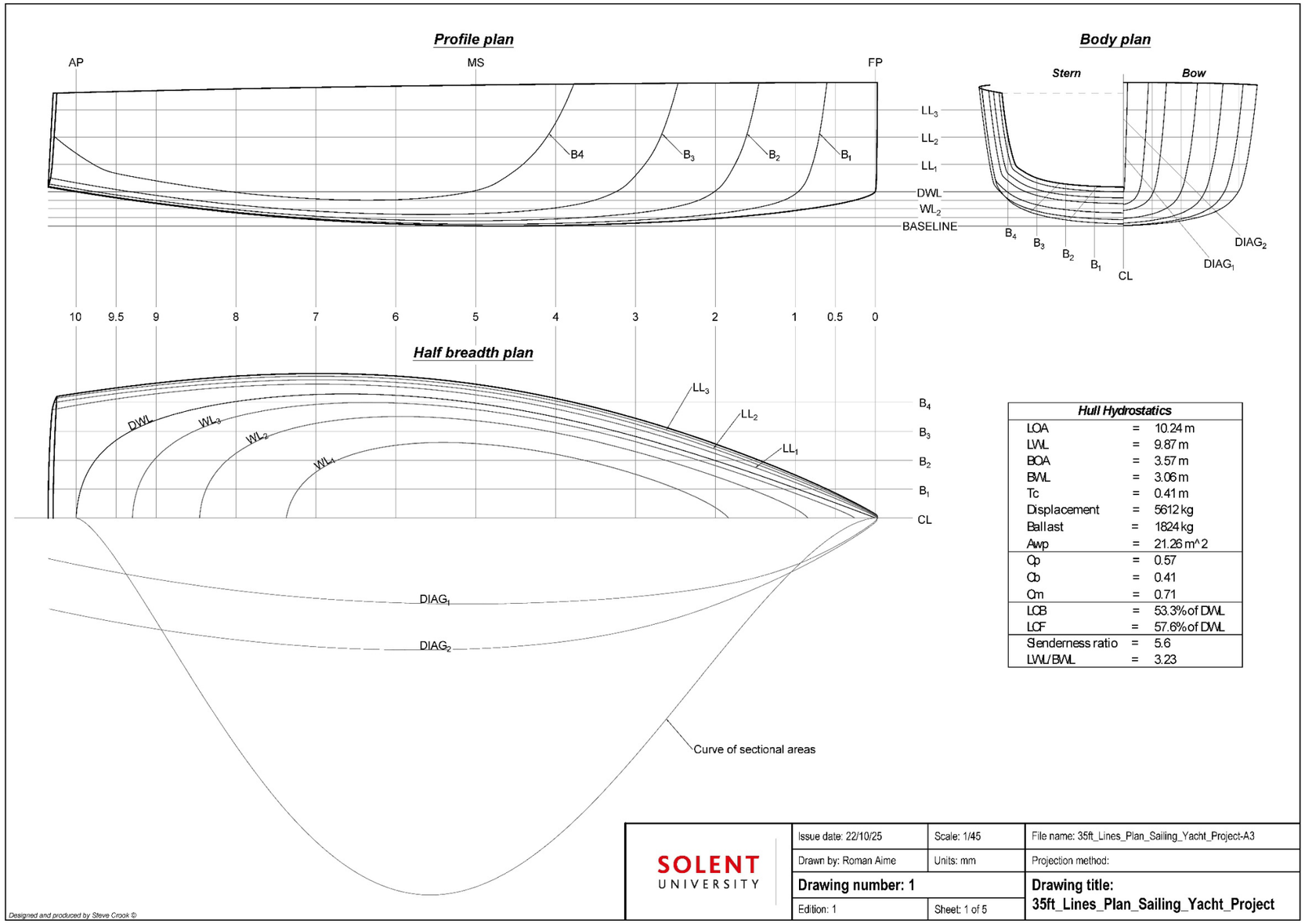
HULL HYDROSTATICS

LOA	10.24 m
LWL	9.87 m
BOA	3.57 m
BWL	3.06 m
Tc	0.41 m
Displacement	5,612 kg
Ballast	1,824 kg
Awp	21.26 m²
Cp	0.57
Cb	0.41
Cm	0.71
LCB	53.3% of DWL
LCF	57.6% of DWL
Slenderness Ratio	5.6
LWL / BWL	3.23

02

Lines Plan

Profile, Half Breadth & Body Plan — Scale 1:45 — Drawing No. 1



Hull Hydrostatics		
LOA	=	10.24 m
LWL	=	9.87 m
BOA	=	3.57 m
BWL	=	3.06 m
Tc	=	0.41 m
Displacement	=	5612 kg
Ballast	=	1824 kg
Awp	=	21.26 m^2
Cp	=	0.57
Cb	=	0.41
Cm	=	0.71
LCB	=	53.3% of DWL
LCF	=	57.6% of DWL
Slenderness ratio	=	5.6
LWL/BWL	=	3.23

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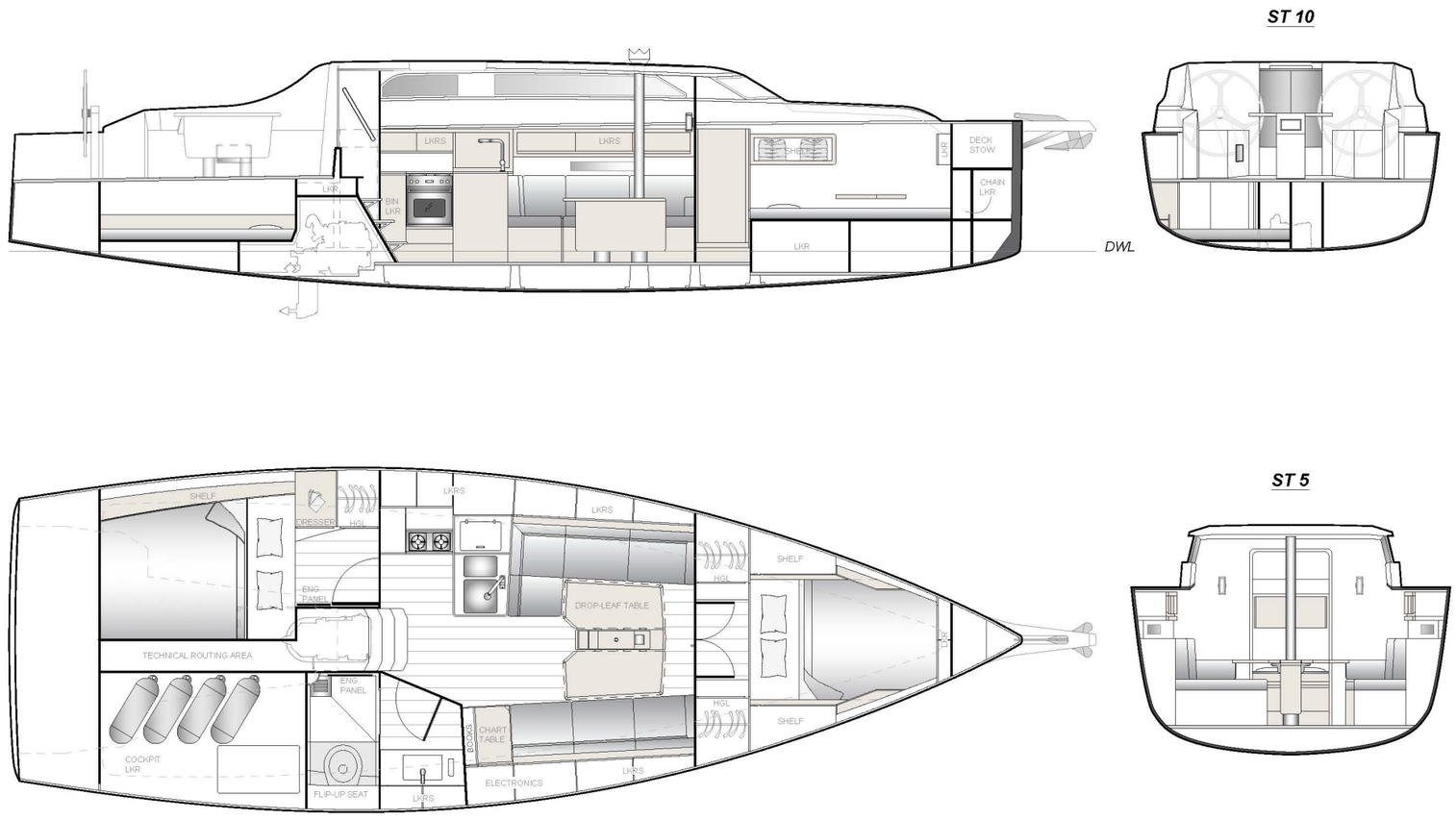
Issue date: 22/10/25	Scale: 1/45	File name: 35ft_Lines_Plan_Sailing_Yacht_Project-A3
Drawn by: Roman Aime	Units: mm	Projection method:
Drawing number: 1		**Drawing title: 35ft_Lines_Plan_Sailing_Yacht_Project**
Edition: 1	Sheet: 1 of 5	

03

General Arrangement

Interior Layout & Sections — Scale 1:40 — Drawing No. 2



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	Drawing number: 2		**Drawing title:** 35ft_GA_Sailing_Yacht_Project
	Edition: 1	Sheet: 2 of 5	

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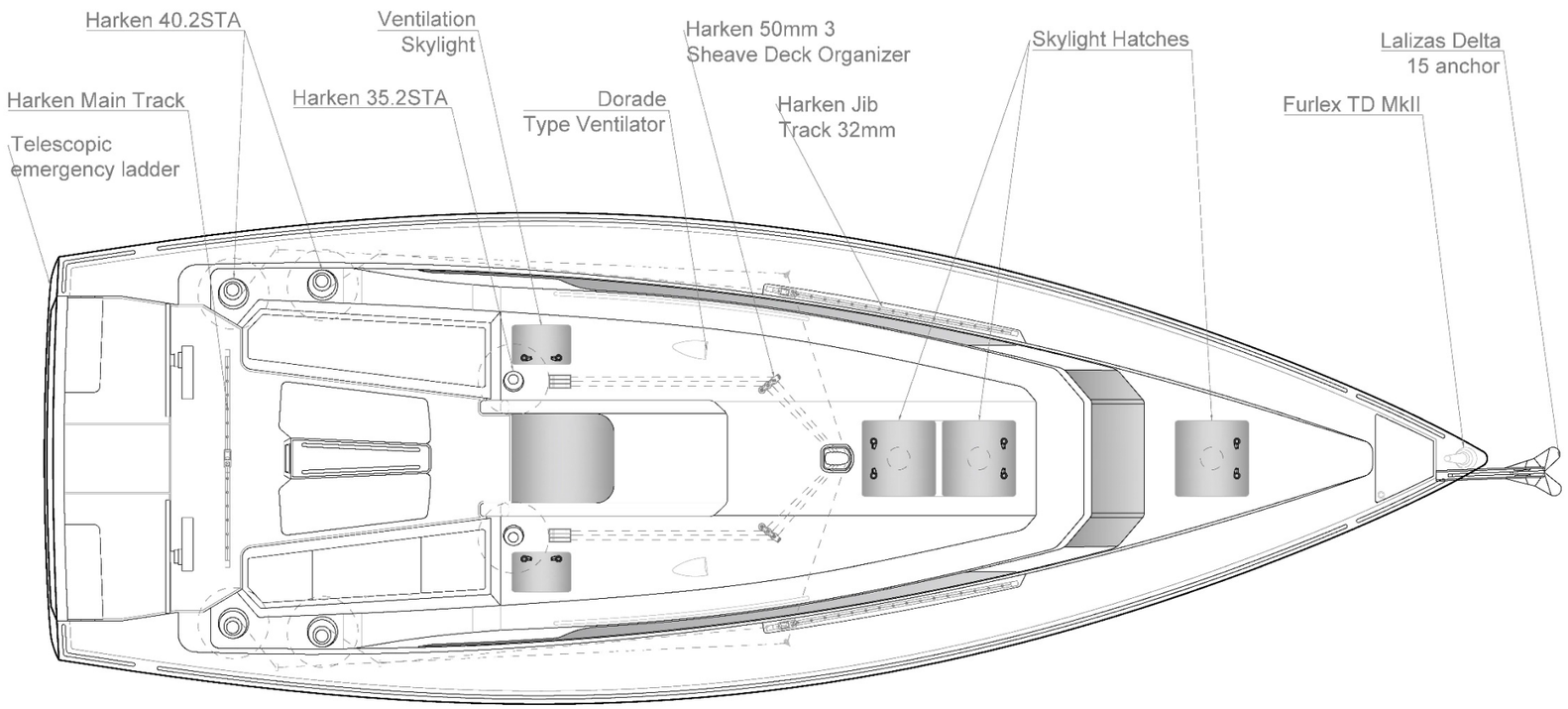
04

Deck Layout

Hardware & Equipment Plan — Scale 1:30 — Drawing No. 3



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	Drawing number: 3		**Drawing title: DECK PLAN MED-35 Q102619070**
	Edition: 1	Sheet: 1 of 1	

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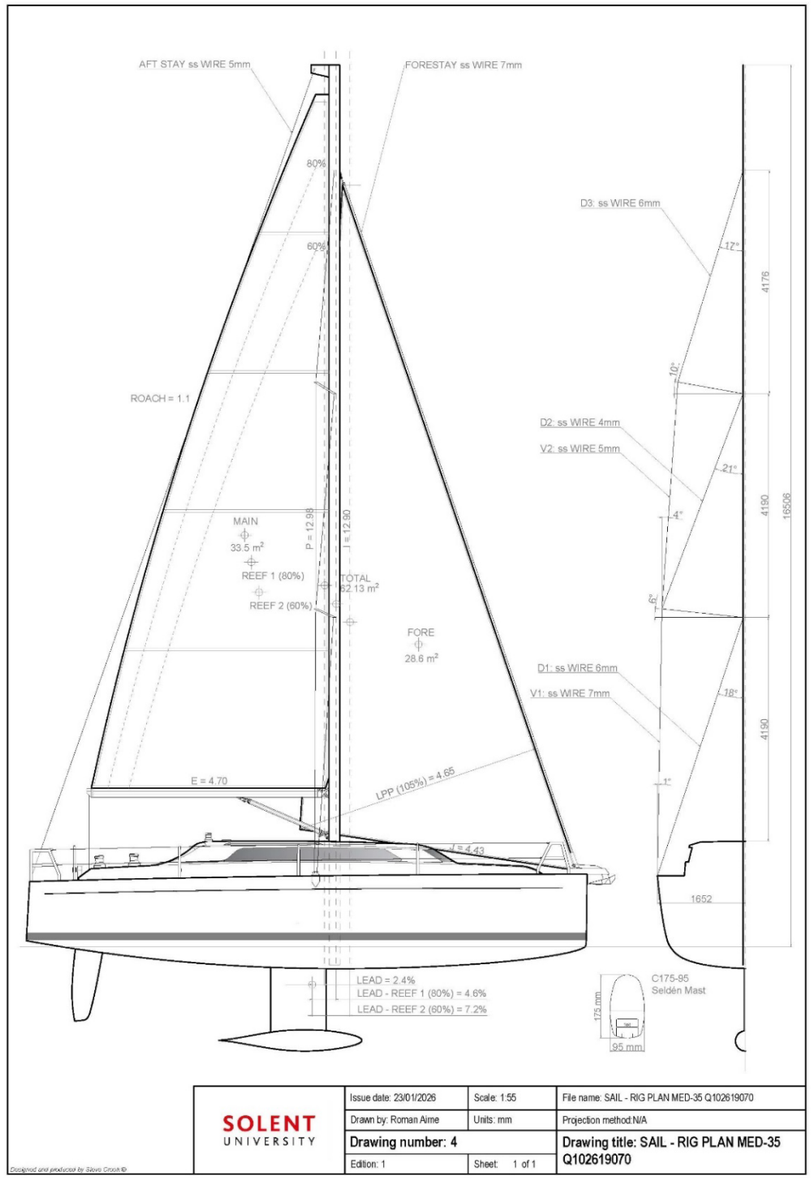
05

Sail & Rig Plan

Fractional Rig, Double Spreaders — Scale 1:55 — Drawing No. 4



SAIL AREAS

Mainsail	33.5 m²	54%
Foresail	28.6 m²	46%
Total Upwind	62.13 m²	
Total (incl. roach)	65.28 m²	

RIG DIMENSIONS

P	12.98 m
E	4.70 m
I (foretriangle height)	12.90 m
J (foretriangle base)	4.43 m
LPP (105%)	4.65 m
Mast Section	Selden C175-95

LEAD & REEFING

Lead	2.4%
Lead — Reef 1 (80%)	4.6%
Lead — Reef 2 (60%)	7.2%

STANDING RIGGING — AISI 316 SS WIRE

Forestay	7 mm
Aft Stay	5 mm
V1	7 mm
D1	6 mm
V2	5 mm
D2	4 mm
D3	6 mm

06

Stability & Appendages

STABILITY ANALYSIS

The stability analysis confirms coherence with the design philosophy. The GZ curve shows a maximum righting arm of 0.62 m at 68.2 degrees of heel, indicating a significant reserve of stability at higher heel angles. The vanishing angle is estimated slightly above 115 degrees, satisfying the required stability standards while remaining dependent on loading condition.

DELLENBAUGH ANGLE

The Dellenbaugh angle calculation yields 14 degrees, within the expected range of 12 to 17.5 degrees. This confirms coherence between the sail plan and initial stability, consistent with the design philosophy of a balanced cruiser-racer.

KEEL DESIGN

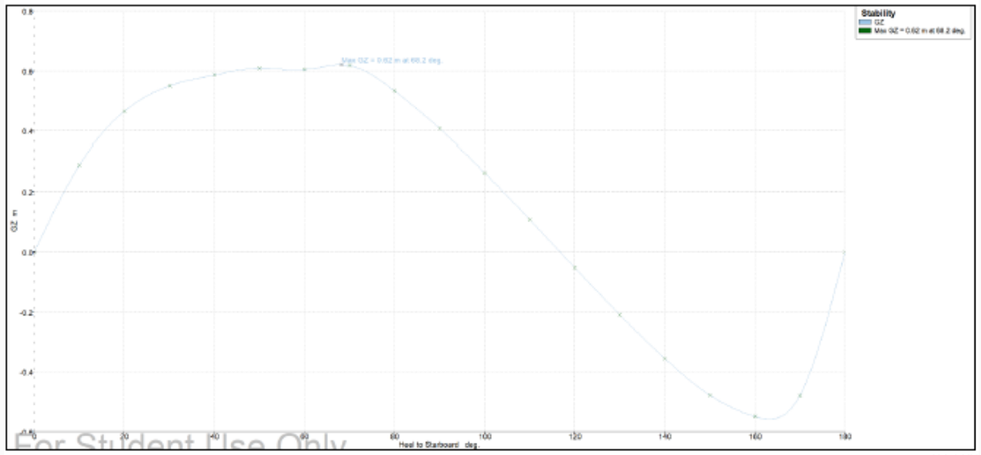
The keel follows a fast cruiser philosophy with cast iron construction for durability and impact resistance. A 70/30 mass distribution between bulb (NACA 65-015) and fin (NACA 64-010) concentrates ballast low, optimising the righting arm. The aspect ratio of 1.46 provides improved lateral efficiency.

RUDDER CONFIGURATION

Twin RUD29 rudders from Jefa ensure effective manoeuvrability when heeled. The rudder area to sail area ratio of 1.6% is consistent for a cruiser-racer of this size.

KEY FIGURES

Max GZ	0.62 m
GZ at angle	68.2°
Vanishing angle	> 115°
Dellenbaugh angle	14°
Leeway angle	3.85°
Keel area / SA	2.55%
Keel AR	1.46
Keel chord	1.040 m
Keel span	1.521 m
Bulb profile	NACA 65-015
Fin profile	NACA 64-010
Rudder area / SA	1.6%
Rudders	2 × Jefa RUD29
Total rudder area	0.99 m²
GM	1.884 m
KG	0.035 m
KB	0.28 m
BM	2.069 m



GZ Curve — Heel to Starboard

For Student Use Only

07

Weight Estimation & Systems

WEIGHT SUMMARY BY GROUP

01 Structure	1,191 kg	21.4%
02 Forepeak	28 kg	0.5%
03 Forward Cabin	110 kg	2.0%
04 Saloon	122 kg	2.2%
05 Navigation Station	60 kg	1.1%
06 Galley	101 kg	1.8%
07 Head	96 kg	1.7%
08 Aft Cabin	80 kg	1.4%
09 Cockpit Stow	60 kg	1.1%
10 Installations	368 kg	6.6%
11 Deck Equipment	216 kg	3.9%
12 Rig & Sails	184 kg	3.3%
13 Ballast (Keel + Bulb)	1,824 kg	32.5%
14 Payload Condition	1,128 kg	20.1%
TOTAL	**5,559 kg**	**100%**

ENGINE & SYSTEMS

Engine
Yanmar 3YM30AE

Battery — Start
90 Ah AGM

Battery — Service
100 Ah Lithium

Fuel Tank
Near VCG, aft of engine

Fresh Water Tank
Under saloon sole

Black Water Tank
Under saloon sole

INTERIOR ARRANGEMENT

2 cabins (fwd + aft port)
1 head (stbd of companionway)
Saloon with drop-leaf table
Seating for 6
L-shaped galley to port
Chart table with electronics
2 m headroom amidships

08

Conclusion

The MED-35 project delivers a coherent compromise between performance and usability, adapted to family sailing in the Mediterranean. Efficient in light wind conditions while remaining controllable and safe as conditions strengthen, the design achieves its core objectives across all measured parameters.

The stability analysis confirms this coherence: a maximum GZ of 0.62 m at 68.2 degrees, with a vanishing angle above 115 degrees, demonstrates a level of stability consistent with a fast cruiser philosophy without moving towards an excessively stiff configuration.

LOA	DISPLACEMENT	SAIL AREA	AVS	BALLAST RATIO
10.24 m	5,612 kg	62.13 m²	> 115°	0.33